Odette School of Business

BSMM- 8730: Data Acquisition and Management

D.4.2 Project Design Document

Global Child Mortality Visual Analytics Application (2010–2024)

Group Members

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1. Introduction & Context

The Global Child Mortality Visual Analytics Application is designed to support the systematic analysis of world's child mortality patterns between 2010 and 2024. The application will integrate mortality data with relevant healthcare indicators to examine temporal trends, differences across countries and geographical regions, and potential relationships between mortality outcomes and healthcare conditions. To support a reliable analytical workflow, Microsoft Excel will be used for the initial examination and basic validation of raw datasets, while R/RStudio will provide the primary environment for data cleaning, transformation, integration, exploratory data analysis, and statistical analysis. MySQL will be used to store and manage the cleaned and integrated datasets, and Microsoft Power BI will provide interactive dashboards, charts, tables, maps, filters, and data-exploration capabilities. The resulting analytical environment will facilitate evidence-based interpretation of mortality patterns and provide findings that can support education, research, and public-health analysis. This design is based on the D.4.1 User Requirements and Analysis Document, including functional requirements F1–F11, non-functional requirements NF1–NF9, and acceptance criteria AC-01–AC-11.

2. Goals & non-goals

The main objective of the Global Child Mortality Visual Analytics Application is to analyze child mortality trends between 2010 and 2024 and provide users with a clear understanding of how mortality rates have changed over time. The application will enable users to compare mortality rates across different countries and geographical regions, while also exploring relationships between child mortality and selected healthcare indicators. Through interactive dashboards, users will be able to filter and examine information by country, region, and year, allowing them to investigate specific patterns and trends more effectively. Exploratory data analysis will be used to identify important variations, relationships, and changes within the data. The application will present these findings through clear tables, charts, maps, and other interactive visualizations that can support education, academic research, and public-health analysis. In addition, users will be able to export selected data and analytical results for further examination, reporting, or research purposes.

Non-goals

The application will not include individual patient records, real-time mortality monitoring, medical diagnosis, or establishing causal relationships between healthcare indicators and child mortality.

Success Criteria

- Dashboard filters should update within 3 seconds under defined testing conditions.
- Historical mortality data from 2010–2024 should be displayed correctly where data is available.
- The world map and charts should update according to the selected filters.
- Cleaned and integrated datasets should be stored and retrieved correctly from MySQL.

3. System Architecture

The application will use a multi-stage data analytics architecture that separates data examination, processing, storage, and visualization. Raw mortality, healthcare, and other relevant datasets will first be examined and validated using Microsoft Excel. The validated data will then be transferred to R/RStudio for cleaning, transformation, standardization, integration, exploratory data analysis, and statistical analysis. The resulting cleaned and integrated dataset will be stored and managed in MySQL. Microsoft Power BI will connect to the analytical data to provide interactive dashboards, charts, tables, maps, and filters for users.

4. Logical Architecture

The logical architecture describes how the main functional components of the Global Child Mortality Visual Analytics application interact, without focusing on the physical hardware or specific deployment environment. The application will follow a layered data-processing architecture in which each component performs a specific role.

Data Source Layer: This layer contains the raw mortality data, healthcare indicators, and other relevant demographic, socioeconomic, or contextual data required for the analysis.

Data Examination and Validation Layer: Microsoft Excel will provide the initial examination of the raw datasets. This layer will support basic validation, missing-value checks, duplicate checks, structural checks, and consistency checks before further processing.

Data Processing and Analysis Layer: R/RStudio will perform the main data-processing activities, including cleaning, transformation, standardization, integration, exploratory data analysis, and statistical analysis.

Data Storage Layer: MySQL will store and manage the cleaned and integrated analytical data. This layer provides structured access to country, region, year, mortality, and healthcare-indicator information.

Visualization and Presentation Layer: Microsoft Power BI will use the processed data to provide interactive dashboards, charts, tables, maps, Key Performance Indicator-KPIs, and filters. Users will be able to examine mortality trends, compare countries and regions, and explore relationships between healthcare indicators and mortality.

User Layer: Users will interact with the Power BI dashboards to explore, interpret, and export selected information for education, research, and public-health analysis.

5. Physical Architecture

This physical architecture describes the main technologies and components used to implement the application:

Data Sources: Mortality data, healthcare indicators, and other relevant socioeconomic, demographic, or contextual data.

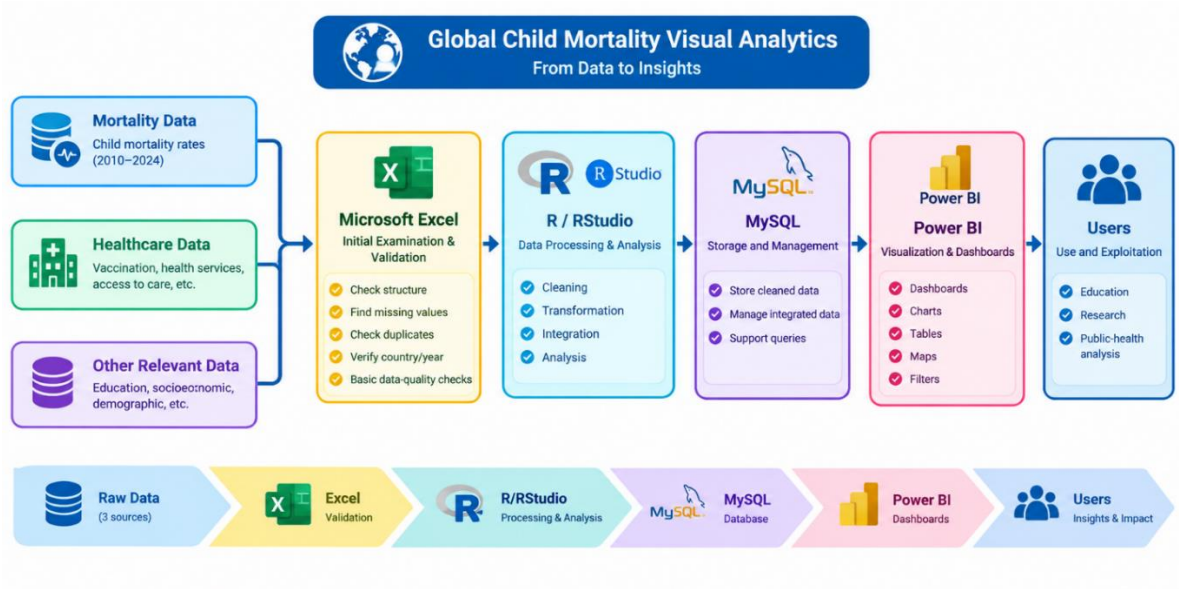
Microsoft Excel: Used on the user's computer for initial examination, preliminary validation, and basic data-quality checks.

R/RStudio: Used as the primary data-processing environment for cleaning, transformation, standardization, integration, exploratory data analysis, and statistical analysis.

MySQL Database: Used to store and manage the cleaned and integrated datasets in a structured database environment.

Microsoft Power BI: Used to connect to the analytical data and provide interactive dashboards, visualizations, filters, tables, charts, and maps.

Users: Researchers, students, educators, and public-health users who interact with the Power BI dashboards and analytical results.

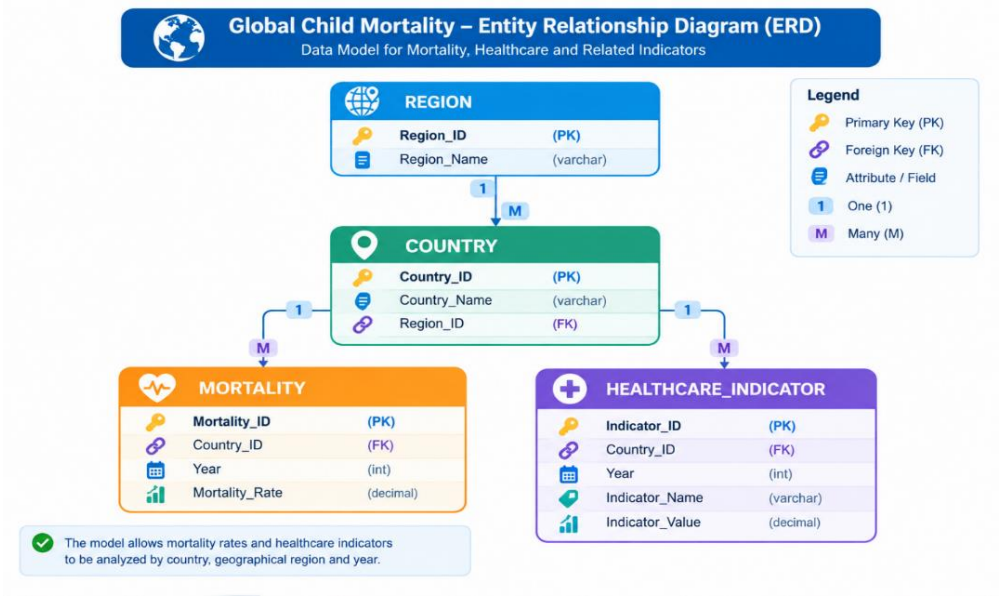


6. Data Model:

The data model will organize the mortality and related indicator data around country, region, year, mortality, and healthcare indicators. A country will belong to a geographical region, while mortality and healthcare measurements will be associated with a specific country and year. The schema includes the following main tables, whose Primary keys will uniquely identify records, while foreign keys will establish relationships between countries, regions, mortality data, and healthcare indicators.

Table	**Purpose**
Region	Stores geographical regions
Country	Stores country information and regional relationships
Mortality	Stores country-year mortality measurements
Healthcare_Indicator	Stores healthcare indicator measurements by country and year

Entity-relationship diagrams-ERD



7. Component Design

Overview – For a quick understanding of the overall situation before exploring more detailed results, the overview page will provide a general picture of global child mortality. It includes Total countries analyzed, Selected year, Global mortality rate, Mortality trend, regional comparison, and Global map.

Trends - Users will examine mortality changes over time. Main visual: Line chart showing mortality from 2010–2024, where the Users can select a country or region to examine its trend.

Country and Region Comparison - This page will allow users to compare Countries, Regions, Mortality rates, Changes over time. Main visuals: Bar chart, Map, Comparison table.

Healthcare Analysis - This page will explore relationships between mortality and healthcare indicators. Main visual: Scatter plot. Users can select different healthcare indicators and years.

Data Explorer - Users will be able to view the underlying data in table form and export selected information.

Findings and Impact - The Findings and Impact section will present analytical results for education, research, and public-health interpretation. Healthier Children will highlight mortality reductions and healthcare improvements. Stronger Communities will emphasize country and regional patterns. A Brighter Future will summarize key evidence supporting research, informed decisions, and public-health planning.

Data Cleaning and Integration Algorithm

BEGIN

Load mortality and healthcare datasets

Check for missing values and duplicate records

Standardize country names, years, and data formats

Integrate compatible datasets

Validate the cleaned data

Store the integrated data in MySQL

END

8. Interface Design

The dashboard interface will feature clear titles, simple navigation, consistent fonts and formatting, clearly labelled charts, easy-to-use filters, and minimal visual clutter.

Country, region, and indicator names will also be presented consistently throughout the application to ensure clarity and a uniform user experience.



The dashboard design will include clear wireframes and screen layouts to define the structure, navigation, filters, charts, tables, maps, and other user-interface components. Where required, API or data-access definitions will describe how Power BI connects to and retrieves data from the analytical database. The implementation will also consider constraints and security, including available hardware and system resources, data-processing and dashboard performance, database capacity, network limitations, user access controls, data protection, and the security of stored and transmitted

data. The interface will remain simple, responsive, and efficient while maintaining appropriate security and data integrity.

9. Design Rationale

This design document defines the requirements, expected outcomes, look and feel, and technical approach for an integrated visual analytics solution for examining global child mortality data. Microsoft Excel was selected for initial data examination and validation because it supports data-structure review and basic quality checks. R/RStudio was selected for data cleaning, transformation, integration, exploratory data analysis, and statistical analysis. MySQL was selected to provide structured storage and efficient management of cleaned and integrated datasets. Microsoft Power BI was selected to deliver interactive dashboards, charts, tables, maps, and filters for exploring mortality patterns and comparisons. Together, these technologies provide a logical workflow—Excel → R/RStudio → MySQL → Power BI → Users—supporting mortality trend analysis, country and regional comparisons, healthcare-relationship analysis, pattern identification, analytical findings, and selected-data export for education, research, and public-health analysis.